Latané Bullock

Auditory & speech signal processing: speech-in-noise, acoustics, hearing science
Cambridge, MA | Available Summer 2027 | latanebullock@gmail.com | +1 817 995 2626
latanebullock.info | github.com/prlabu | linkedin.com/in/latanebullock

EDUCATION

HARVARD UNIVERSITY
Ph.D., Speech & Hearing Bioscience and Technology
Expected Fall 2027 | GPA 4.0

RICE UNIVERSITY
B.S. Elec. & Computer Eng.
B.A. Linguistics
2020 | GPA 3.87
Exchange: Nat'l Univ. of Singapore

Coursework: Sound, Speech & Hearing (MIT HST.714), Biology of the Inner Ear, Anatomy of Speech & Hearing, Clinical Aspects of Speech & Hearing, Random Processes

SKILLS

Audio & Acoustics
Transfer-function measurement, inverse filtering, LPC & formant analysis, spectrogram & cochleogram features, speech in babble noise, voice acoustics, Praat

Signal Processing & ML
Filter design, FFT & Hilbert analysis, time warping, clustering, SVM, GMM-HMM, permutation statistics

Programming
Python (NumPy, SciPy, scikit-learn, MNE), MATLAB (Signal Processing, App Designer, FieldTrip), R (tidyverse, ggplot2), Git, Bash/Unix
Familiar: PyTorch, GPU cluster jobs, AI coding agents (Claude Code), C, JavaScript

Languages
English (native), French & Spanish (conversational), Portuguese (basic)

HONORS

NIH F31 NRSA Fellowship (PI), 2024
NIH T32 Training Grant, 2021
FENS Brain Conference Stipend, 2026
Delgutte Scholarship, Harvard SHBT, 2026
Fulbright Netherlands Semifinalist, 2019

EXPERIENCE

HARVARD MEDICAL SCHOOL | Ph.D. Researcher, MGH Brain Modulation Lab & BU Speech Neuroscience Lab
2022 – Present | Boston, MA

- Found that the basal ganglia track **ambient noise level separately from speech output**, a candidate neural basis of the Lombard effect (23 patients, 90 LFP sites, 83 single units); first-author **Nature Communications**.
- Showed that auditory cortex **compresses its dynamic range under multitalker babble**, while motor cortex stays invariant to vocal intensity.
- Co-led **SEAM-Stress**, a multilevel prosody model mapping phoneme sequences to auditory and articulatory targets; **synthesized speech from articulography** with self-supervised models (WavLM, SPARC).
- Won an **NIH F31 fellowship** as PI; mentored 2 graduate students; taught the program's stats & ML bootcamp; 15+ talks, including an invited talk in China.

JOHNS HOPKINS JSALT WORKSHOP | Undergraduate Researcher
Summer 2019 | Montreal, Canada

- Developed **overlap-aware speaker diarization** (GMM-HMM resegmentation with neural overlapped-speech detection), reaching **state-of-the-art on AMI, DIHARD, ETAPE**; first-author **ICASSP**.

UTHEALTH MCGOVERN MEDICAL SCHOOL | Research Engineering Assistant, Tandon Lab
May 2020 – May 2021 | Houston, TX

- Analyzed sEEG/ECoG from **115 patients and 8,000+ trials**; showed supplementary motor area activity first predicts speech reaction time; first-author **iScience**.

PROJECTS

3D-PRINTED VOCAL TRACTS 2021 | GitHub

- Measured acoustic transfer functions of 3D-printed vocal tracts; designed an **inverse filter** to flatten the speaker–coupler response; showed a tube model reproduces **F1–F3 within 100 Hz** (FFT, LPC).

TUG OF WORDS 2019 – 2020 | Rice University

- Built hardware and software counting caregiver–child conversational turns via **speaker diarization**, with a web dashboard for parents.

SELECTED PUBLICATIONS

9 peer-reviewed papers (4 first-author) | full list at latanebullock.info

- **Bullock L** et al. **Independent basal ganglia neural populations encode speech production and ambient-noise levels.** Nature Communications, 2026.
- Zhu Y, …, **Bullock L**. **Human subthalamic neurons encode speech features during listening and couple with auditory cortex.** Nature Communications, in press.
- **Bullock L** et al. **Acoustic-aerodynamic voice outcome ratios identify changes in vocal function following vocal fold medialization.** J. Voice, 2023.